MIL-C-915/39C
20 August 1973
SUPERSEDING[1/]
MIL-C-915/39B
5 April 1973

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL TYPE PI

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:
Qualification required.

Construction

First - Copper conductor, nickel coated, Navy Standard size 2(7).
Second - Extruded silicone rubber insulation. Nominal thickness 0.018-inch.
Third - Glass braid (one colored black and one colored white for each pair of conductors).
Fourth - Two conductors (one black, one white), cabled together with a lay not greater than 2-1/4 inches, to form a pair.
Fifth - A braided shield of AWG No. 34 or No. 36 uncoated copper over each pair. Minimum braid angle 30 degrees; minimum coverage 85 percent.
Sixth - Shield insulation over each braided shield, consisting of polyethylene terephthalate film, type G of MIL-I-631, 0.001-inch thick, applied helically with 50 percent overlap.
Seventh - A glass braid over each shield insulation. (Standard Identification Code, applied by method 4).
Eighth - The specified number of pairs (see details) cabled together with a lay not greater than 14 inches. Asbestos fillers saturated with non-charring silicone-base compound shall be used as necessary to make a firm, round-cable.
Ninth - Binder tape applied helically with overlap.
Tenth - Silicone rubber jacket (see details for thickness).
Eleventh - Braided metal armor.
Twelfth - Paint.

Details

Type and Size	Number of pairs	Cable jacket thickness, minimum, (inch)	Overall diameter, maximum, (inch)	Cold bending mandrel diameter, maximum (inch)
PI-3	3	0.040	0.685	9.0
PI-7	7	.050	.900	11.0
PI-12	12	.050	1.155	14.0

EXAMINATION AND TESTS:

	Requirements:
Basic electrical:	
Conductor resistance - Ohms/1000 ft. at 25°C., maximum[2/]	6.9
Voltage withstand - Volts, rms, minimum	
Conductor to conductor	1200
Conductor to shield	700
Shield to Shield	500
Insulation resistance - Megohms/1000 ft., minimum	500

1/ Requirements formerly covered on MIL-C-23206/2.
2/ Resistance shall be measured on 100 percent of conductors; values shall be included in a certified test report.

© Denotes changes.

FSC 6145

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Group A:	Requirements:
Visual and dimensional	
Group B:	
Cold bending, cable - (See details for mandrel diameter)	No damage
Gas flame, 1 hour at 120/70 volts, rms.	No failure
Physicals (unaged)	
Insulation	
Tensile strength - psi, minimum	750
Elongation - Percent, minimum	125
Jacket	
Tensile strength - psi, minimum	800
Elongation - Percent, minimum	250
Durometer (Type A) hardness, minimum (on specimen of jacket) (ASTM-D 2240)	60
Group C:	
Physicals - Cable aging (260° ± 10°C.)	
Jacket	
Tensile strength - psi, minimum	600
Elongation - Percent, minimum	150

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical plus all of Groups A, B and C plus the following:

Cable aging and compatibility (125° ± 5°C.) - (For this test, omit all references to watertightness and fillers)	No failure
Gas flame (3 hours) at 120/70 volts, rms.	No failure
Armor. Conformance to material, construction, coverage	
Specific gravity of extruded silicone insulation, maximum	1.55

UNIT ORDERING LENGTHS:

All types and sizes . . . . . 100 feet, minimum.

Custodians:
Army - MI
Navy - SH
Air Force - 80

Preparing activity:
Navy - SH
(Project 6145-0656)

Review activities:
Army - EL, MI, AV
Navy - EC
Air Force - 80

User activities:
Army - ME, MU, WC
Navy - CG

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED S/N 0102-014-1802

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